Iterators in Python -

What is an iteration

Iteration is a general term for taking each item of something, one after another, Any time you use a loop, explicit or implicit to go over a group of items, that is iteration.

for loop - iteration over sequence

Sequence - (ordered collections)
1. String ✓
2. list ✓
3. tuple ✓
4. dictionary ✓
5. range ✓

ordered

①
```
name = "akshay"
for i in name:
```

print(i)

o/p - a
k
s
h
a
y

② list1 = [10, 20, 30, 40, 50]

```
for i in list1:
    print(i)
```

o/p - 10
20
30
40
50

③ tuple1 = (10, 20, 30, 40, 50)

```
for i in tuple1:
    print(i)
```

o/p - 10
20
30
40

50

④ dict1 = { "name" : "akshay",
"roll_no" : 25 }

```
for key,value in dict1.items():
    print(key,value)
```

o/p - name akshay
roll_no 25

⑤ for i in range(1,11,1):
(1,2,3 ... 10)
print(i)

o/p -
1
2
3
:
10

- what is iterator ?

- An iterator is an object that allows the programmer to traverse through a

sequence of data without having to store the entire data in the memory.

- we perform iteration through iterator.

list1 = [10, 20, 30, 40, 50]

```
for i in list1:
    print(i)
```

What is iterable ?

Iterable is an object, which one can iterate over.

It generates an iterator when passed to iter() method.

list2 & var is an obj. we can

elements one by one, and it is called as iterable

```
list1 = [1, 2, 3, 4, 5]
print(type(list1))  ⟶ list

for i in list1:
    print(i)
```

o/p- 1 2 3 4 5

~~iteration~~
~~iterable~~
iterator

This list1 is an iterable.

```
iter(list2)  ⟶ <list iterator at 0x21>
```

⇓

```
print(type(iter(list2)))  ⟶ list_iterator.
```

Built-in fun -

len()

```
list1 : [10, 20, 30]
print(len(list1))  ⟶ 3
```

method -

append

list1 = [10, 20, 30]

list1.append(40) → [10, 20, 30, 40]

How to identify what is iterable & what is iterator.

1. every iterable has an iter function.

2. every iterator has both iter function as well as next function.

dir(list2) →

dir(var) →